

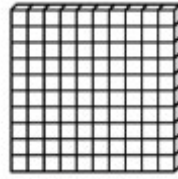
OKLAHOMA SCHOOL TESTING PROGRAM

TEST BLUEPRINT AND ITEM SPECIFICATIONS
GRADE 5 MATHEMATICS

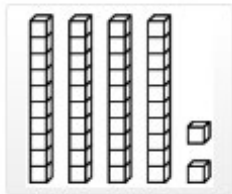
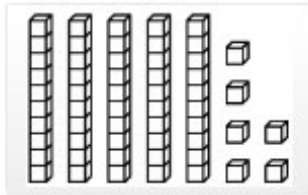


OKLAHOMA
Education

This model represents 1.



Match the model in the left column to the correct fraction in the right column. Each model in the left column matches to only one fraction in the right column. Click one model on the left and then click its match on the right. To remove a connection, hold the pointer over the line until it turns red, and then click it.



$$\frac{1}{10}$$

$$\frac{56}{100}$$

$$\frac{42}{100}$$

$$\frac{4}{10}$$

Standard: 5.N.1.1 Represent decimal fractions (e.g., $\frac{1}{10}$, $\frac{1}{100}$) using a variety of models (e.g., 10 by 10 grids, base-ten blocks, meter stick) and show the rational number relationships among fractions, decimals and whole numbers.

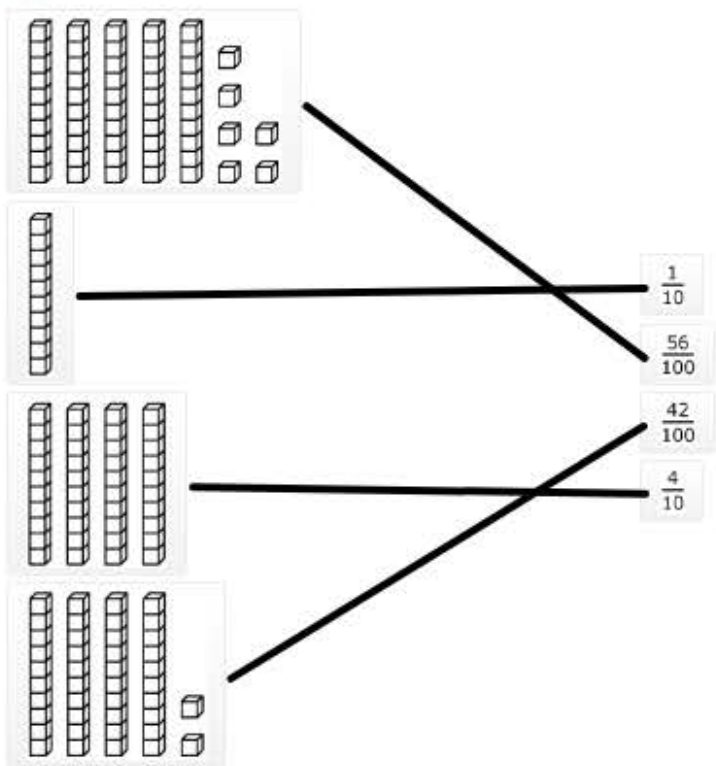
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, identifying a model that represents a fraction.

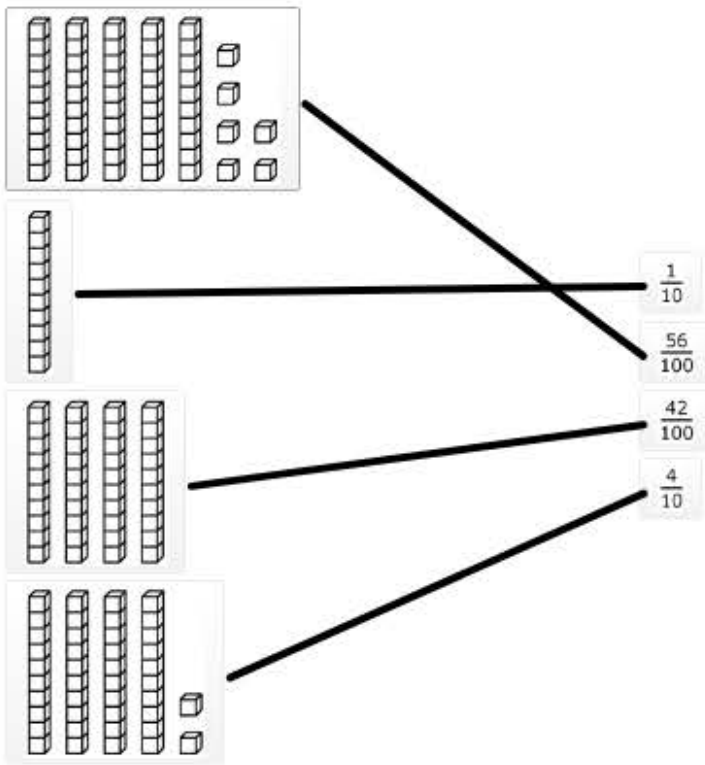
1 continued...

Distractor Rationale:

Correct

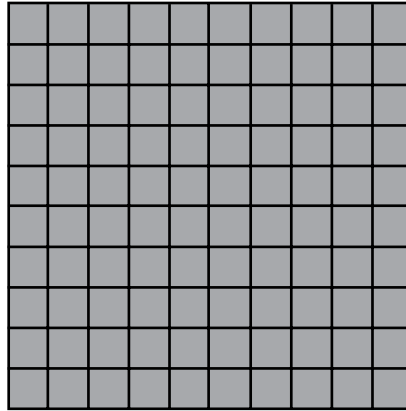


Incorrect

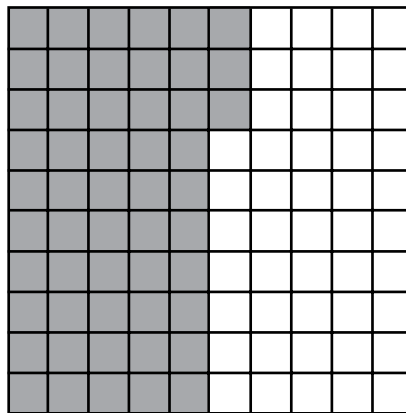


The student ignored the 2 ones and just saw the 4 tens.

- 2** This model represents 1 whole.



The shaded part of this model represents a number.



What number is represented by the shaded part of this model?

- A** 0.053
- B** 0.53
- C** 5.30
- D** 50.3

Standard: 5.N.1.2 Read, write, and represent decimals using place value to describe decimal numbers including fractional numbers as small as thousandths and whole numbers up to seven digits.

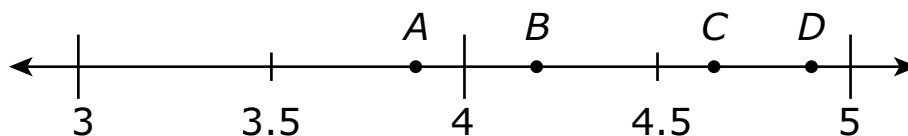
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform the simple procedure of identifying a decimal given a model.

Distractor Rationale:

- A. The student made a place value error.
- B. Correct. The student demonstrated an ability to identify the decimal represented by a model.**
- C. The student made a place value error.
- D. The student saw 5 whole rods and 3 units.

- 3** Which point on the number line below best represents the location of 4.82?



- A** point A
- B** point B
- C** point C
- D** point D

Standard: 5.N.1.3 Compare and order decimals and fractions, including mixed numbers and fractions less than one, and locate on a number line.

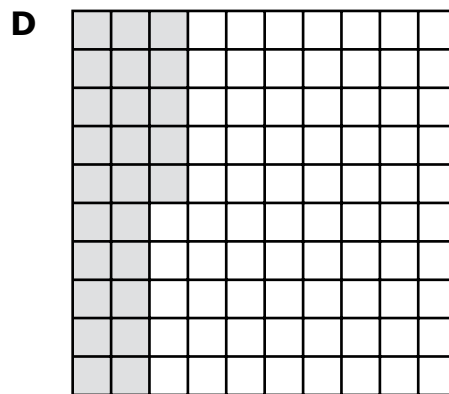
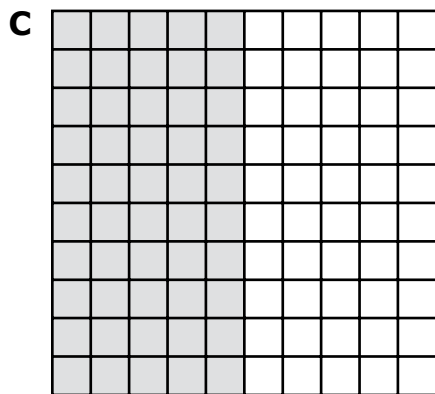
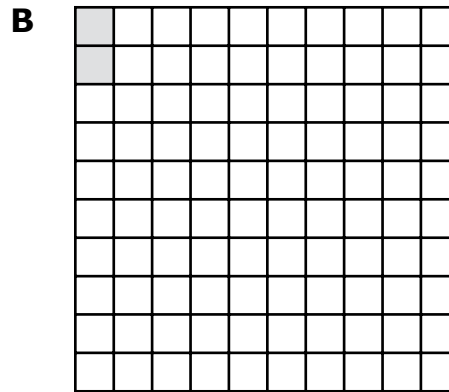
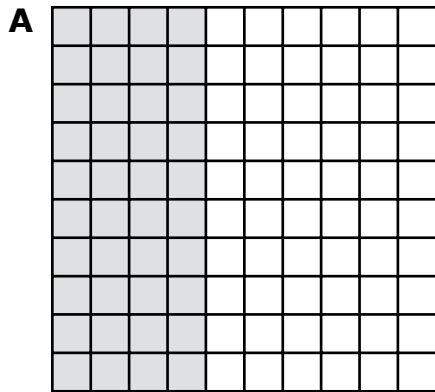
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to first know that 4.82 is between 4.5 and 5. The student must then further determine if it is closer to 4.5 or 5 to choose the correct answer.

Distractor Rationale:

- A. The student thought 4.82 is to the left of 4.
- B. The student confused 4.82 and 4.2.
- C. The student knew that 4.82 is more than 4.5, but did not go far enough.
- D. **Correct.** The student demonstrated an ability to locate a decimal on the number line.

- 4 In Tara's class, $\frac{2}{5}$ of the students had cereal for breakfast. Which grid has $\frac{2}{5}$ of its area shaded?



Standard: 5.N.1.4 Recognize and generate equivalent terminating decimals, fractions, mixed numbers, and fractions in various models.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to translate $\frac{2}{5}$ into a model out of 100.

Distractor Rationale:

- A. Correct. The student demonstrated an ability to recognize equivalent decimals, as shown on a hundred grid, and fractions.
- B. The student focused on the numerator.
- C. The student focused on the denominator.
- D. The student thought $\frac{2}{5}$ was the same as $\frac{25}{100}$.

5 Which decimal number means the same as $\frac{12}{100}$?

- A** 0.012
- B** 0.12
- C** 1.2
- D** 12

Standard: 5.N.1.4 Recognize and generate equivalent terminating decimals, fractions, mixed numbers, and fractions in various models.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, turning a decimal fraction into a decimal.

Distractor Rationale:

- A. The student confused hundredths and thousandths.
- B. Correct. The student demonstrated an ability to generate an equivalent decimal to a fraction.**
- C. The student confused hundredths and tenths.
- D. The student used the numerator as a whole number.

- 6** An art teacher has 381 beads for making necklaces. Students will put 18 beads on each necklace.
- Which is the **best** estimate for the number of necklaces the students can make?
- A** 10 necklaces
 - B** 20 necklaces
 - C** 30 necklaces
 - D** 40 necklaces

Standard: 5.N.2.1 Estimate solutions to division problems to assess the reasonableness of results.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to interpret the scenario and then find the best estimate.

Distractor Rationale:

- A. The student underestimated the number of necklaces.
- B. Correct. The student demonstrated an ability to estimate the solution to a division problem.**
- C. The student overestimated the number of necklaces.
- D. The student overestimated the number of necklaces.

- 7** The art teacher had 450 sheets of construction paper on the first school day. The art students used 18 sheets during each school day. How many school days did the construction paper last?
- A** 21
 - B** 22
 - C** 24
 - D** 25

Standard: 5.N.2.2 Divide multi-digit numbers, by one- and two-digit divisors, based on knowledge of place value, including but not limited to standard algorithms.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine what operation to use before completing the division.

Distractor Rationale:

- A. The student made a computational error.
- B. The student rounded 18 to 20, then chose the closest value.
- C. The student made a computational error.
- D. Correct. The student demonstrated an ability to divide a multi-digit number by a two-digit divisor.**

- 8** A student sorted 950 crayons into boxes that hold 36 crayons each. How many more crayons will the student need to completely fill the last box?
- A** 6
 - B** 14
 - C** 22
 - D** 30

Standard: 5.N.2.2 Divide multi-digit numbers, by one- and two-digit divisors, based on knowledge of place value, including but not limited to standard algorithms.

Depth-of-Knowledge: 3

This item is a DOK 3 because it requires the student to solve a non-routine problem. The student must first find the amount that will be left over and then decide how to use this to determine the number needed to completely fill the last box.

Distractor Rationale:

- A. The student focused on the ones place of the quotient.
- B. The student identified the remainder after dividing.
- C. Correct. The student demonstrated an ability to solve a real-world problem requiring division of multi-digit whole numbers.**
- D. The student estimated the answer by dividing 900 by 30.

9 Todd bought a coat on sale for \$29.95. He gave the clerk \$40. Which is closest to the amount of change he should have received?

- A** \$60
- B** \$20
- C** \$15
- D** \$10

Standard: 5.N.3.1 Estimate sums and differences of fractions with like and unlike denominators, mixed numbers, and decimals to assess the reasonableness of the results.

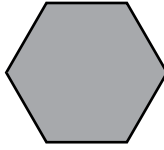
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine how to solve the problem and then find the closest, not the exact, answer.

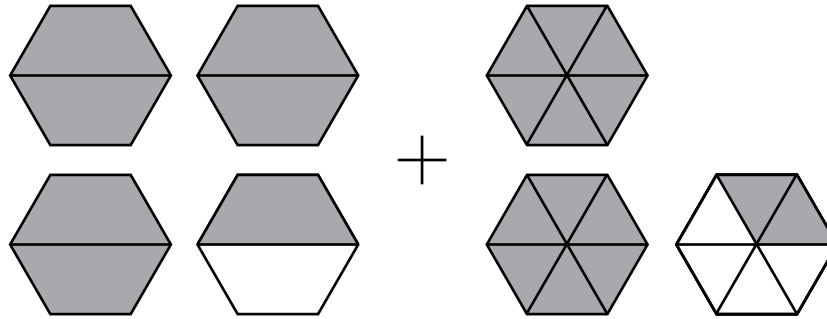
Distractor Rationale:

- A. The student rounded \$29.95 to \$20 and then added that to \$40.
- B. The student rounded \$29.95 to \$20.
- C. The student computed the actual amount of change and then rounded the answer up to the next largest answer choice, \$15.
- D. **Correct.** The student demonstrated an ability to estimate a difference with decimals.

- 10** The model represents one whole.



This model shows an addition expression.



Which fraction is equivalent to the sum of the shaded parts of this model?

- A** $1\frac{1}{6}$
- B** $1\frac{2}{6}$
- C** $5\frac{3}{6}$
- D** $5\frac{5}{6}$

Standard: 5.N.3.2 Illustrate addition and subtraction of fractions with like and unlike denominators, mixed numbers, and decimals using a variety of mathematical models (e.g., fraction strips, area models, number lines, fraction rods).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to interpret the model and then compute.

Distractor Rationale:

- A. The student subtracted instead of added.
- B. Balance distractor.
- C. The student found 5 wholes and 3 parts, ignoring that the parts were not the same size.
- D. **Correct.** The student demonstrated an ability to add mixed numbers with unlike denominators given a model.

11 Anna's assignment is to read for a total of 5 hours (hr) in two days. She read $2\frac{1}{4}$ hours on the first day. How many hours does Anna need to read on the second day to complete her assignment?

A $2\frac{1}{4}$ hr

B $2\frac{3}{4}$ hr

C $3\frac{1}{4}$ hr

D $3\frac{3}{4}$ hr

Standard: 5.N.3.3 Add and subtract fractions with like and unlike denominators, mixed numbers, and decimals, involving money, measurement, geometry, and data. Use various models and efficient strategies, including but not limited to standard algorithms.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine how to approach the problem when subtracting a mixed number from a whole number.

Distractor Rationale:

- A. The student thought she would read the same amount each day or confused $2\frac{1}{4}$ with $2\frac{1}{2}$.
- B. Correct. The student demonstrated an ability to subtract a mixed number from a whole number.**
- C. The student subtracted the whole numbers and then thought the fraction would still be $\frac{1}{4}$.
- D. The student made a regrouping error when subtracting.

12 Misty created the number pattern below.

32, 28, 24, 20, . . .

If n represents a number in this pattern, which rule could be used to find the next number in the pattern?

- A** $n + 4$
- B** $n - 4$
- C** $n \cdot 4$
- D** $n \div 4$

Standard: 5.A.1.1 Use tables and rules of up to two operations to describe patterns of change and make predictions and generalizations about various mathematical situations.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to first figure out the pattern and then explain the relationship between terms in the pattern using an algebraic expression.

Distractor Rationale:

- A. The student chose an incorrect operation.
- B. Correct. The student demonstrated an ability to describe a pattern of change with a rule.**
- C. The student chose an incorrect operation.
- D. The student chose an incorrect operation.

- 13** Nick started a chess club. The table shows the total number of members at the end of each month.

Chess Club

Month	Total Members
1	2
2	5
3	9
4	14
5	20

Based on the pattern shown in the table, how many total members were in the club at the end of month 7?

- A** 27 members
- B** 32 members
- C** 35 members
- D** 54 members

Standard: 5.A.1.1 Use tables and rules of up to two operations to describe patterns of change and make predictions and generalizations about various mathematical situations.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to first determine the pattern and then use this rule to apply it to month 7.

Distractor Rationale:

- A. The student found the total number of members who were in the club at the end of month 6.
- B. The student saw that there were 6 more members from month 4 to 5 and thinks that there will be 6 more members each for months 6 and 7.
- C. Correct.** The student demonstrated an ability to use a table to describe a pattern of change.
- D. The student computed $14 + 20$ and added that to the 20 members in the club at the end of month 5.

14 Which expression is equivalent to $(4 \times 8) + (4 \times 3)$?

- A** $4 + (8 \times 3)$
- B** $4 + (8 + 3)$
- C** $4 \times (8 \times 3)$
- D** $4 \times (8 + 3)$

Standard: 5.A.2.1 Generate equivalent numerical expressions and solve problems using number sense involving whole numbers by applying the commutative property, associative property, distributive property, and order of operations (excluding exponents).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to apply the distributive property to identify equivalent expressions.

Distractor Rationale:

- A. The student swapped multiplication and addition.
- B. The student thought addition and multiplication are interchangeable.
- C. The student thought addition and multiplication are interchangeable.
- D. Correct. The student demonstrated an ability to generate equivalent numerical expressions involving whole numbers using the distributive property.**

- 15** Elise handed out paper to 8 students. Each student received 3 pieces of yellow paper, 4 pieces of red paper, and 2 pieces of black paper. Elise wrote this expression to show the total number of pieces of paper she handed out.

$$8 \times (3 + 4 + 2)$$

Which is equivalent to the expression Elise wrote?

- A** $(8 \times 3) + (4 + 2)$
- B** $(8 + 3) + (8 + 4) + (8 + 2)$
- C** $(8 + 3) \times (8 + 4) \times (8 + 2)$
- D** $(8 \times 3) + (8 \times 4) + (8 \times 2)$

Standard: 5.A.2.1 Generate equivalent numerical expressions and solve problems using number sense involving whole numbers by applying the commutative property, associative property, distributive property, and order of operations (excluding exponents).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to apply the distributive property to identify equivalent expressions.

Distractor Rationale:

- A. The student did not realize that the multiplication sign needs to apply to the 4 and 2 as well as the 3.
- B. The student did not understand how to apply the multiplication sign before the addition expression.
- C. The student did not understand how to apply the multiplication sign before the addition expression.
- D. Correct. The student demonstrated an ability to generate equivalent numerical expressions by applying the distributive property.**

16 An expression is shown.

$$15 - 2x$$

What is the value of this expression when $x = 4$?

- A** 7
- B** 9
- C** 23
- D** 52

Standard: 5.A.2.3 Evaluate expressions involving variables when values for the variables are given.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to substitute 4 for x and then find the value of the expression.

Distractor Rationale:

- A. Correct. The student demonstrated an ability to evaluate an expression involving a variable when a value for the variable is given.
- B. The student computed $15 - (2 + 4)$ instead of $15 - (2 \times 4)$.
- C. The student added instead of subtracted.
- D. The student selected because both 5 and 2 are in the expression.

- 17** Triangle PQR has at least two angles that are congruent.
All of the following could describe triangle PQR **except** which term?
- A** right
 - B** scalene
 - C** isosceles
 - D** equilateral

Standard: 5.GM.1.1 Describe, identify, classify, and construct triangles (equilateral, right, scalene, isosceles) by their attributes using various mathematical models.

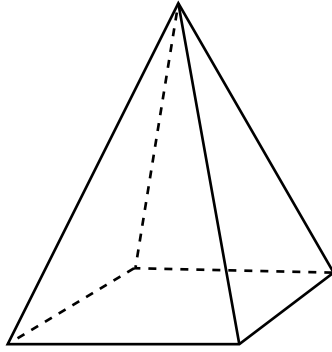
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use an understanding of the characteristics of triangles to determine which triangles are possible with the given information.

Distractor Rationale:

- A. The student did not realize that right triangles can have two congruent angles.
- B. Correct. The student demonstrated an understanding of the attributes of specific triangles.**
- C. The student did not understand the attributes of an isosceles triangle.
- D. The student did not understand the “at least” part of the triangle’s description.

18 Wade drew a square pyramid for his art project.



Which statement is true of the square pyramid?

- A** The square pyramid has 8 faces.
- B** The square pyramid has 2 bases.
- C** The square pyramid has 5 vertices.
- D** The square pyramid has a triangular base.

Standard: 5.GM.1.2 Describe, identify, and classify three-dimensional figures (cubes, rectangular prisms, and pyramids) and their attributes (number of edges, faces, vertices, shapes of faces), given various mathematical models.

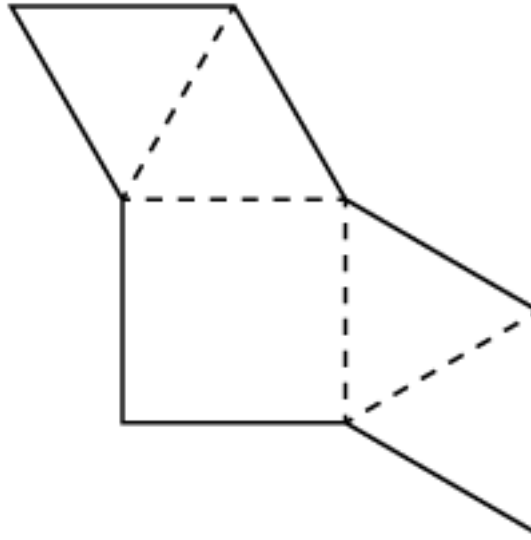
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to recall facts and definitions related to a square pyramid.

Distractor Rationale:

- A. The student confused faces and edges.
- B. The student confused square pyramids and square prisms (cubes).
- C. Correct. The student demonstrated an ability to describe a square pyramid by the number of edges, faces, or vertices as well as the shape of faces.**
- D. The student confused base and face.

19 A net of a three-dimensional figure is shown.



Which three-dimensional figure does this net represent?

- A** cube
- B** square pyramid
- C** rectangular prism
- D** triangular pyramid

Standard: 5.GM.1.3 Recognize and draw a net for a three-dimensional figure (cube, rectangular prism, pyramid).

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to recall the fact of what the net of a square pyramid looks like.

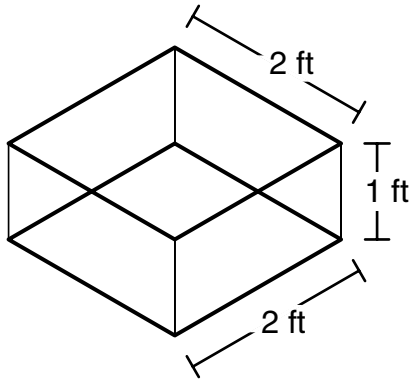
Distractor Rationale:

- A. The student focused only on the square base.
- B. Correct. The student demonstrated an ability to recognize a net for a three-dimensional figure.**
- C. The student focused only on the rectangular base.
- D. The student focused only on the triangular faces.

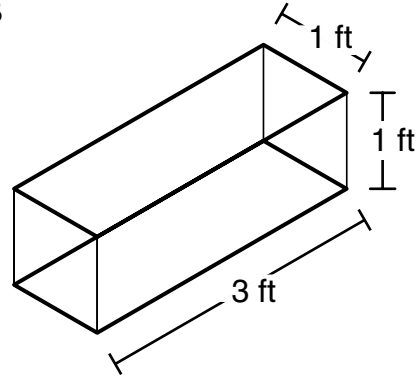
Pedro wants to buy an aquarium in the shape of a rectangular prism for his fish. He will choose from four options.

Which aquarium will have the greatest volume?

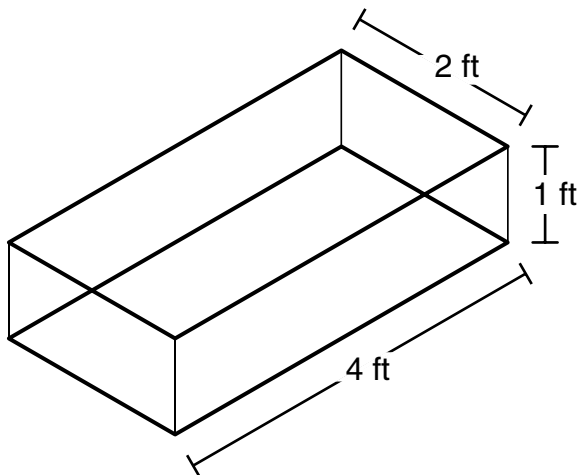
A



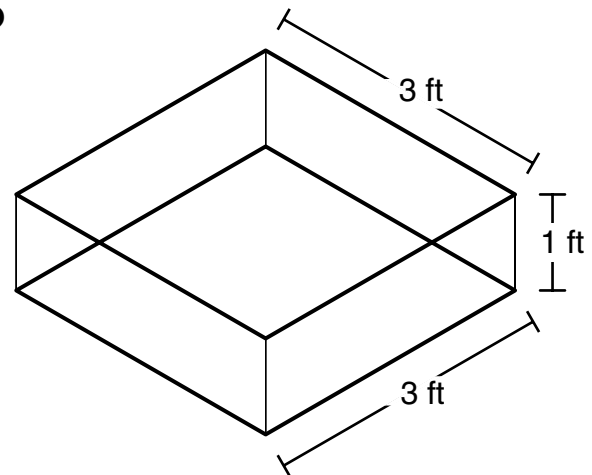
B



C



D



Standard: 5.GM.2.1 Determine the volume of rectangular prisms by the number of unit cubes (n) used to construct the shape and by the product of the dimensions of the prism $a \cdot b \cdot c = n$. Understand rectangular prisms of different dimensions (p , q , and r) can have the same volume if $a \cdot b \cdot c = p \cdot q \cdot r = n$.

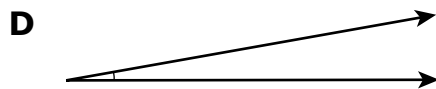
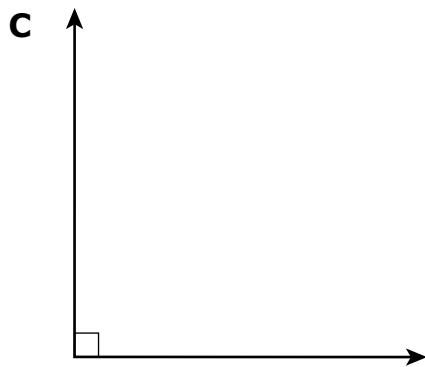
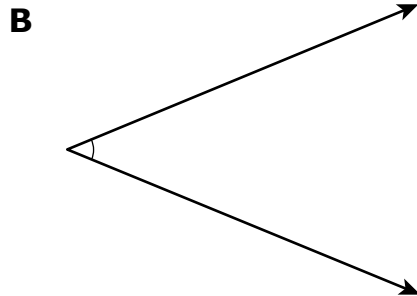
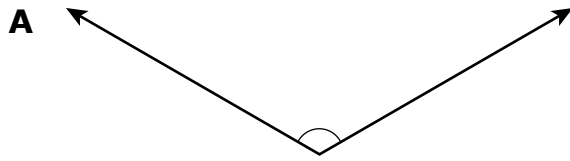
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to find the volume of four rectangular prisms and then determine the one with the largest volume.

Distractor Rationale:

- A. The student made a multiplication error.
- B. The student made a multiplication error or thought Pedro wanted the aquarium with the least volume.
- C. The student chose the aquarium with the greatest single dimension.
- D. **Correct.** The student demonstrated an ability to find the volume of rectangular prisms.

21 Which angle has the greatest measure?



Standard: 5.GM.3.1 Measure and compare angles according to size using various tools.

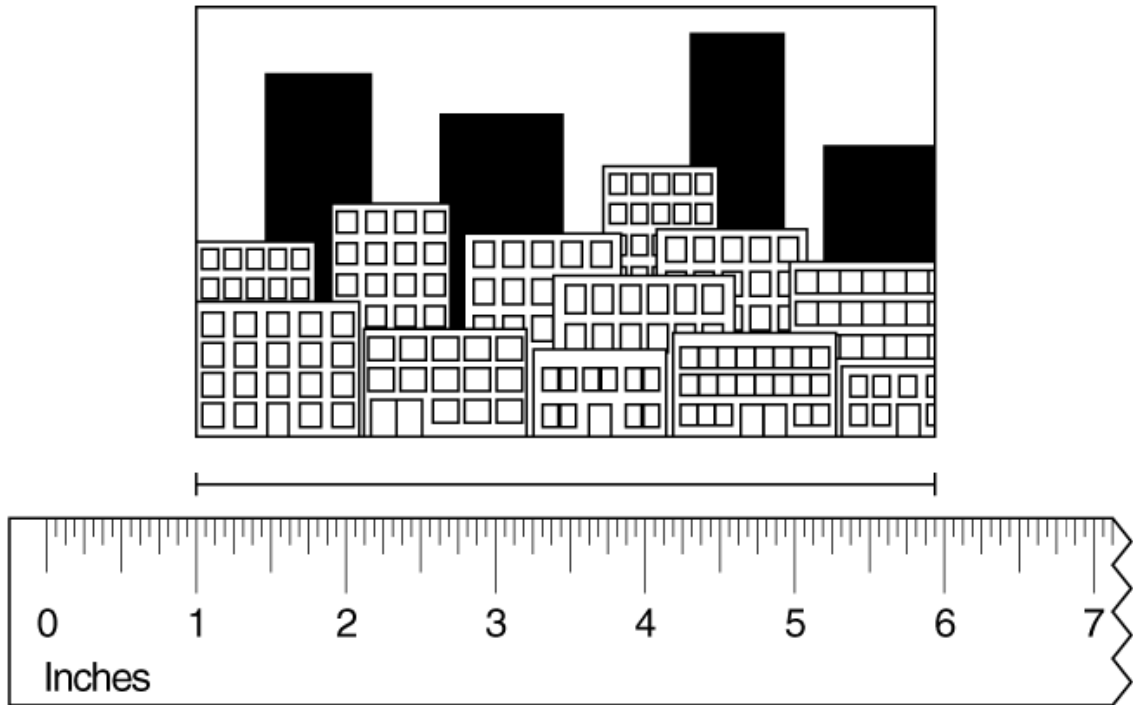
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to compare angle measures and find the one with the greatest measure.

Distractor Rationale:

- A. **Correct.** The student demonstrated an ability to compare angles according to size.
- B. Balance distractor
- C. The student thought right angles are always largest.
- D. The student confused largest and smallest.

22 A picture of a city is shown.



What is the length of this picture to the nearest $\frac{1}{16}$ of an inch?

- A** $4\frac{1}{16}$ inches
- B** $4\frac{15}{16}$ inches
- C** $5\frac{1}{16}$ inches
- D** $5\frac{15}{16}$ inches

Standard: 5.GM.3.2 Measure the length of an object to the nearest whole centimeter or up to $\frac{1}{16}$ -inch using an appropriate instrument.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, measuring using a ruler.

Distractor Rationale:

A. Balance distractor

B. Correct. The student demonstrated an ability to measure the length of an object to the nearest $\frac{1}{16}$ -inch.

C. The student confused $\frac{1}{16}$ less than 5 inches with $\frac{1}{16}$ greater than 5 inches.

D. The student did not take into account that the picture begins at 1 inch, not 0 inches.

- 23** An aisle at a supermarket has a length of 24 yards. What is the length of this aisle in feet?
- A** 2 feet
 - B** 8 feet
 - C** 72 feet
 - D** 288 feet

Standard: 5.GM.3.3 Apply the relationship between inches, feet, and yards to measure, convert, and compare objects to solve problems.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use a strategy to convert from one unit of measure to another.

Distractor Rationale:

- A. The student computed 24 divided by 12.
- B. The student computed 24 divided by 3.
- C. Correct. The student demonstrated an ability to use the relationship between yards and feet.**
- D. The student computed 24 multiplied by 12.

24 The heights, in centimeters, of some bean plants are listed.

14, 19, 19, 26, 27, 32, 32, 32, 33

What is the median height of these bean plants?

- A** 19 centimeters
- B** 26 centimeters
- C** 27 centimeters
- D** 32 centimeters

Standard: 5.D.1.1 Find the measures of central tendency (i.e., mean, median, mode) and range of a set of data. Understand that the mean is a “leveling out” or central balance point of the data.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to recall the definition of median.

Distractor Rationale:

- A. The student confused median and mode and did not see that 32 is actually the mode.
- B. The student confused median and mean.
- C. **Correct.** The student demonstrated an ability to find the median of a set of data.
- D. The student confused median and mode.

- 25** This table shows the high temperatures for some Oklahoma towns in January 2010.

High Temperatures

Town	Temperature (°F)
Arnett	67
Beaver	70
Boise City	64
Buffalo	68
Goodwell	67
Kenton	64
Slapout	70

What is the range of these high temperatures?

- A** 2°
- B** 3°
- C** 4°
- D** 6°

Standard: 5.D.1.1 Find the measures of central tendency (i.e., mean, median, mode) and range of a set of data. Understand that the mean is a “leveling out” or central balance point of the data.

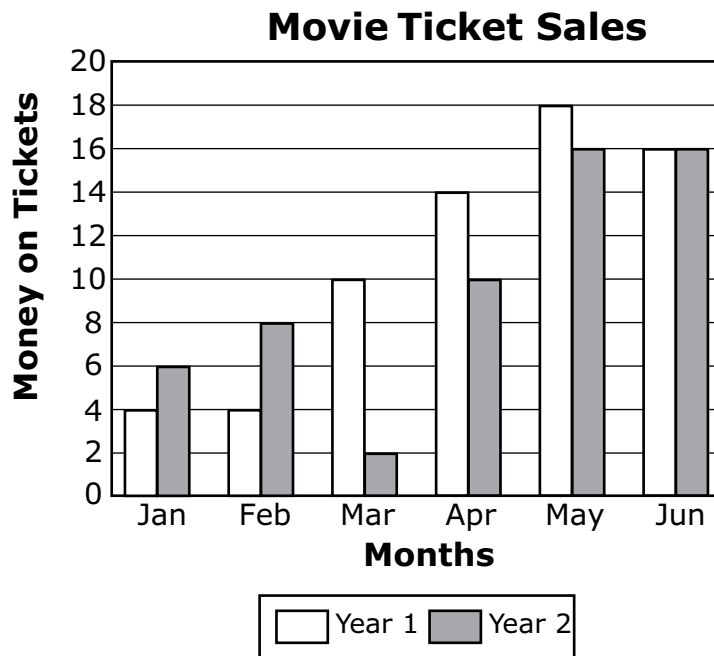
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to identify the high and low temperatures in a table and then use these to calculate the range.

Distractor Rationale:

- A. The student found the different between the highest temperature and the second highest temperature.
- B. The student found the range for the first and last temperatures in the table.
- C. Balance distractor
- D. **Correct.** The student demonstrated an ability to find the range of a set of data.

- 26** The graph below shows how much Tisha spent on movie tickets for the first six months of two different years.



Which month had the greatest difference in the amount of money spent on movie tickets between year 1 and year 2?

- A** March
- B** April
- C** May
- D** June

Standard: 5.D.1.2 Create and analyze line and double-bar graphs with increments of whole numbers, fractions, and decimals.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use a double bar graph to answer a question about the data.

Distractor Rationale:

- A. Correct.** The student demonstrated an ability to analyze a double-bar graph with whole numbers.
- B. Balance distractor**
- C.** The student chose the month with the highest bar in the graph.
- D.** The student did not know what difference means and identified two bars with the same height in the graph.

Cluster Items

The following sample items are part of a cluster. The cluster is presented first and then the two items that follow require use of the cluster. The two items are from different standards.

Use the information to answer the following questions.

A baker made 64 doughnuts at his bakery on Monday.

27 The baker continues to make the same number of doughnuts each day. How many days will it take to make an additional 768 doughnuts?

- A** 10 days
- B** 12 days
- C** 100 days
- D** 120 days

Standard: 5.N.2.2 Divide multi-digit numbers, by one- and two-digit divisors, based on knowledge of place value, including but not limited to standard algorithms.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine the appropriate operation and then perform division with a two-digit divisor.

Distractor Rationale:

- A. The student made a division error.
- B. Correct. The student demonstrated an ability to divide a multi-digit number by two-digit divisor.**
- C. The student made division and place value errors.
- D. The student made a place value error.

28 The baker wants to put his doughnuts into boxes. Each box holds 12 doughnuts. What is the total number of boxes the baker can fill, and the total number of doughnuts he will have left over?

- A** 8 boxes with 6 doughnuts left over
- B** 6 boxes with 8 doughnuts left over
- C** 5 boxes with 4 doughnuts left over
- D** 4 boxes with 5 doughnuts left over

Standard: 5.N.2.3 Recognize that remainders can be represented in a variety of ways, including a whole number, fraction, or decimal. Determine the most meaningful form of a remainder based on the context of the problem.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to complete a division problem and then make a decision about what to do with the remainder.

Distractor Rationale:

- A. Balance distractor
- B. The student computed $6 \times 12 = 72$ and then thought the 8 as left over meant subtraction.
- C. Correct. The student demonstrated an ability to represent a quotient as a number and an amount left over.**
- D. The student confused the number of boxes and the number left over.